

1. When solving an equation, the goal is to isolate the variable.

Two students, James and Joe, were solving the equation $6 - 5(11 - 2x) = 17x$. Their work is shown below for questions 2 – 5.

James' Work	Joe's Work
$6 - 5(11 - 2x) = 17x$	$6 - 5(11 - 2x) = 17x$
$\frac{6 - 5(11 - 2x)}{6} = \frac{17x}{6}$	$6 - 55 + 10x = 17x$
$-5(11 - 2x) = \frac{17x}{6}$	$-49 + 10x = 17x$
	$\underline{-10x \quad -10x}$
$-55 + 10x = \frac{17x}{6}$	$\frac{-49}{7} = \frac{7x}{7}$
$\underline{-10x \quad -10x}$	$-7 = x$
$-55 = -\frac{43x}{6}$	
$\underline{\times 6 \quad \times 6}$	
$-330 = -43x$	
$\frac{330}{43} = x$	

2. Describe the different first steps the students used in solving the equation.
James divided everything by 6. Meanwhile, Joe distributed -5 over the subtraction group, $(11 - 2x)$.
3. Which student made an error?
James made an error.

4. Write a note to the student who made an error explaining the mistake they made.

Answers may vary.

Subtract 6 instead and then divide by -5 .

5. Is 8 a solution to the equation $3(x - 2) + 5 = 23$? Justify your answer mathematically.

Yes.

$$3(8 - 2) + 5 = 23$$

$$3(6) + 5 = 23$$

$$18 + 5 = 23$$

$$23 = 23 \checkmark$$

For questions 6 – 10, solve each equation and verify the solution using substitution.

6. $-5(2r - 0.3) + 0.5(4r + 3) = -64$

$$-10r + 1.5 + 2r + 1.5 = -64$$

$$-8r + 3 = -64$$

$$-8r = -67$$

$$r = 8.375 \text{ or } 8\frac{3}{8}$$

$$-5(2(8.375) - 0.3) + 0.5(4(8.375) + 3) = -64$$

$$-5(16.45) + 0.5(36.5) = -64$$

$$-82.25 + 18.25 = -64$$

$$-64 = -64 \checkmark$$

7. $\frac{3m+5}{3} = 9m + 2$

$$3\left(\frac{3m+5}{3} = 9m + 2\right)$$

$$3m + 5 = 27m + 6$$

$$-1 = 24m$$

$$-\frac{1}{24} = m$$

$$\frac{3\left(-\frac{1}{24}\right)+5}{3} = 9\left(-\frac{1}{24}\right) + 2$$

$$-\frac{1}{24} + \frac{5}{3} = -\frac{9}{24} + 2$$

$$-\frac{1}{24} + \frac{40}{24} = -\frac{9}{24} + \frac{48}{24}$$

$$\frac{39}{24} = \frac{39}{24} \checkmark$$

8. $\frac{2}{3} + 13k - 1 = 3k - \frac{5}{3}$

$$13k - \frac{1}{3} = 3k - \frac{5}{3}$$

$$13k = 3k - \frac{4}{3}$$

$$10k = -\frac{4}{3}$$

$$k = -\frac{2}{15}$$

$$\frac{2}{3} + 13\left(-\frac{2}{15}\right) - 1 = 3\left(-\frac{2}{15}\right) - \frac{5}{3}$$

$$-\frac{26}{15} - \frac{1}{3} = -\frac{6}{15} - \frac{5}{3}$$

$$-\frac{26}{15} - \frac{5}{15} = -\frac{6}{15} - \frac{25}{15}$$

$$-\frac{31}{15} = -\frac{31}{15} \checkmark$$

9. $4.1 - 7.2z + 3z = 5.2 + z$

$$-7.2z + 3z - z = 5.2 - 4.1$$

$$-5.2z = 1.1$$

$$z = -\frac{11}{52}$$

$$4.1 - 7.2\left(-\frac{11}{52}\right) + 3\left(-\frac{11}{52}\right) = 5.2 + \left(-\frac{11}{52}\right)$$

$$\frac{41}{10} + \frac{99}{65} - \frac{33}{52} = \frac{1,297}{260}$$

$$\frac{1,297}{260} = \frac{1,297}{260} \checkmark$$

10. $\frac{1}{4}x - 3(x - 2) = -2.25x + 1$

$$0.25x - 3x + 6 = -2.25x + 1$$

$$-2.75x + 2.25x = 1 - 6$$

$$-0.5x = -5$$

$$x = 10$$

$$\frac{1}{4}(10) - 3(10 - 2) = -2.25(10) + 1$$

$$2.5 - 3(8) = -22.5 + 1$$

$$2.5 - 24 = -21.5$$

$$-21.5 = -21.5 \checkmark$$

Algebra 1
Unit 5
Lesson 2 Homework

Name Answer Key
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1. Solve the equation and verify the solution using substitution.

$$\begin{aligned} -\frac{1}{8}y - 32 &= -\frac{3}{4}(-8 - 3y) & -\frac{1}{8}y + 7 &= 39 - \frac{3}{4}(-8 - 3y) & -\frac{1}{8}(-16) + 7 &= 39 - \frac{3}{4}(-8 - 3(-16)) \\ \frac{1}{6}y + \frac{128}{3} &= -8 - 3y & & & 2 + 7 &= 39 - 30 \\ 304 &= -19y & & & 9 &= 9 \checkmark \\ -16 &= y & & & & \end{aligned}$$

The formula for calculating the slope of a line from the points (x_1, y_1) and (x, y) is given by the following equation for questions 2-3, $m = \frac{y - y_1}{x - x_1}$.

The point-slope form of a line is as follows: $y - y_1 = m(x - x_1)$

2. Which operation (addition, subtraction, multiplication, or division) can be used to rearrange the slope formula into the point-slope form of a line?

multiplication

3. Use your answer from question 2 to rearrange the slope formula to get the point-slope form of a line.

$$\begin{aligned} (x - x_1) \cdot m &= \frac{y - y_1}{x - x_1} \cdot (x - x_1) \\ m(x - x_1) &= y - y_1 \end{aligned}$$

Use the standard form of a line is given by the formula $Ax + By = C$ for questions 4 - 5.

4. Rearrange this formula to solve for the variable, y .

$$By = C - Ax \qquad y = \frac{C - Ax}{B}$$

5. Based on your answer to question 4, what is the slope of the line you found?
How do you know? Explain your thinking in words.

The slope is $-\frac{A}{B}$. This is because the slope of any line, in the form $y = mx + b$, is m , which in this case is $-\frac{A}{B}$ since $y = \frac{-A}{B}x + \frac{C}{B}$.

For questions 6 - 7, rearrange the formulas to isolate the specified variables.

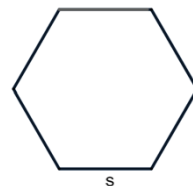
6. The formula $K = \frac{1}{2}mv^2$ is used to find the kinetic energy of an object in motion using its mass, m , and its velocity (speed), v . Solve the equation for the mass, m .

$$2K = mv^2 \qquad \frac{2K}{v^2} = m$$

7. The formula for the volume of a cylinder is given by $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height. Solve the equation for the height, h .

$$\frac{V}{\pi r^2} = \frac{\pi r^2 h}{\pi r^2} \qquad h = \frac{V}{\pi r^2}$$

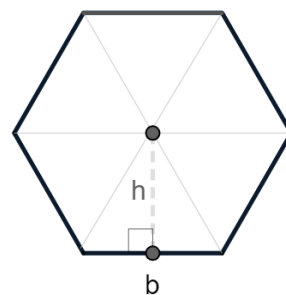
8. The formula for the area of a regular hexagon, A , is $A = \frac{3\sqrt{3}}{2} \cdot s^2$, where s is the side-length.



Lashonda rearranged the area formula to isolate the side-length, s . She knows that the area of the hexagon is 64.95. Her work is shown below. Complete the table to explain each step of Lashonda's work.

Lashonda's work:	Explanation of the steps
$A = \frac{3\sqrt{3}}{2} \cdot s^2$	Formula for the area of a regular hexagon
$2 \cdot A = \cancel{2} \cdot \frac{3\sqrt{3}}{\cancel{2}} \cdot s^2$ $2A = 3\sqrt{3} \cdot s^2$	Multiply both sides of the equation by 2
$\frac{2A}{3\sqrt{3}} = \frac{3\sqrt{3} \cdot s^2}{\cancel{3\sqrt{3}}}$ $\frac{2A}{3\sqrt{3}} = s^2$	Division Property of Equality
$\sqrt{\frac{2A}{3\sqrt{3}}} = \sqrt{s^2}$ $\sqrt{\frac{2A}{3\sqrt{3}}} = s$	Square Root Property of Equality
$\sqrt{\frac{2(64.95)}{3\sqrt{3}}} = s$	Substitution of Known Value
$5 \approx s$	Simplification and Rounding

Xavier is Lashonda's classmate. He said that he did not need the formula for the area of a hexagon to find the side length. He connected the vertices of the hexagon to form 6 equilateral triangles, shown in the figure to the right. He used the formula for the area of a triangle, $A_{\text{triangle}} = \frac{1}{2}bh$, to create the following formula for the area of a regular hexagon: $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$.



Xavier got stuck because he did not know the height, h , of the triangles!

9. Rearrange Xavier's formula, $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$, to solve for the height, h .

$$\frac{A_{\text{hexagon}}}{6} = \frac{6\left(\frac{1}{2}bh\right)}{6}$$

$$\left(\frac{2}{b}\right) \cdot \frac{A_{\text{hexagon}}}{6} = \frac{1}{2}bh \cdot \left(\frac{2}{b}\right)$$

$$\frac{A_{\text{hexagon}}}{3b} = h$$

10. Substitute the area, 64.95, and the side-length Lashonda found, 5, to find the height of Xavier's triangles.

$$\frac{(64.95)}{3(5)} = h$$

$$\frac{64.95}{15} = h$$

$$4.33 = h$$

The heights are equal to 4.33 each.

Algebra 1
Unit 5
Lesson 3 Homework

Name Answer Key
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1. The parts of each equation below are identified by a symbol. Sort the cards into the boxes below by sketching their symbol into the box.

Cards

$$y = -0.2 \cdot x + 3$$

△ ◇ □ ○

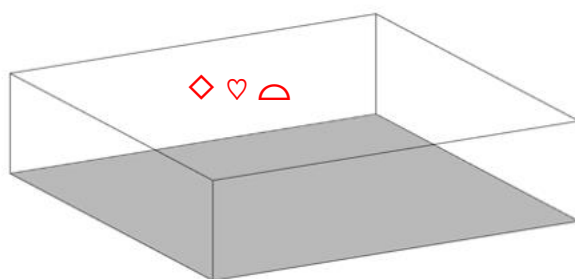
$$11 + -0.2 \cdot m = 9$$

⊖ ♥ ★ ⊙

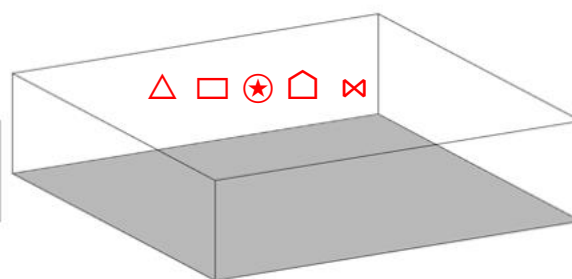
$$y = -0.2 \cdot x + 3$$

⬠ ∩ ✕ ⊠

Boxes



Parameters



Variables

May opened her bank account with money she received as a birthday present. She also deposits the money she saves monthly. The balance, in dollars, in May's bank account after x months can be modeled using the equation $y = 10x + 25$.

2. Which term(s) in the equation are variables?

x and y

3. Which value(s) in the equation are parameters?

10

4. What does the value 25 mean in terms of the balance in May's account?

May's initial balance

5. What does the value 10 mean in terms of the balance in May's account?

The balance in May's account increases by 10 each month.

Choose numbers from the positive digits 1 – 9 to fill in the boxes below for questions 6-10. **Answers will vary.**

$$y = \boxed{5} \cdot x + \boxed{9}$$

6. Create a scenario that your equation could model.
A video posted online currently has 9 views but is now getting 5 views per hour.
7. List the variables in your equation.
 x and y
8. What do each of the variables represent in terms of your scenario?
 **x represents the time, in hours
 y represents the total number of views the video has**
9. List the parameter in your equation.
5
10. What do each of the parameters represent in terms of your scenario?
5 represents the number of views the video receives every hour

A wildlife biologist surveyed the manatee population in Blue Springs State Park in 2009. He observed 326 manatees during that visit. He returned to the park each year since and found that, on average, there were 11 fewer manatees.

1. What are the quantities that are changing in this scenario?

Manatees per year

2. Define the variables m and y .

Answers will vary.

m represents the number of years since 2009.

y represents the number of manatees in Blue Springs State Park.

3. What is the parameter?

-11

4. What is the constant?

326

5. Which equation represents this scenario?

☐ $m = 326y - 11$

☐ $m = -11y + 326$

☒ $y = 326 - 11m$

☐ $y = -11 + 326y$

6. How many manatees were observed in 2015?

$m = 6, \quad y = 326 - 11(6)$

$= 326 - 66$

$= 260 \text{ manatees}$

7. Use the equation to find out how many years it would take for the manatee population to become extinct in Blue Springs if the trend observed by the wildlife biologist continues.

$0 = 326 - 11m$

$-326 = -11m$

$29\frac{7}{11} = m$

It will take approximately 30 years.

In 2021, the Florida Environmental Protection Agency (FL EPA) implemented a new program to feed lettuce to the manatees during the winter to try and stabilize the population. They predict that beginning that year, they can reverse the shrinking trend in the manatee population in Blue Springs so that it will now grow at the rate of 2 manatees per year.

8. Use the equation from the previous question to determine the manatee population in 2021.

$$\begin{aligned} 2021 - 2009 &= 12 = m & y &= 326 - 11(12) \\ & & &= 326 - 132 \\ & & &= 194 \text{ manatees} \end{aligned}$$

9. Write an equation using the variables m and y that represents the new predicted population of manatees after 2021.

$$y = 194 + 2m, m \text{ represents years after 2021}$$

10. Use your equation to predict in what year the manatee population will reach the same size it was in 2009.

$$\begin{aligned} 326 &= 194 + 2m \\ 132 &= 2m \\ 66 &= m \end{aligned}$$

The manatee population will reach 326, 66 years after 2021, in the year 2087.

Marissa and Mykuria are planning a trip to Orlando to celebrate graduation. They want to buy a *three*-day ticket to their favorite theme park, which costs \$299 per person but they also need to find a place to stay during their trip. They found a beautiful vacation rental in Orlando that has room for a number of guests and costs a total of \$871.50. If Marissa and Mykuria are able to fill the room up to its maximum capacity with their friends, the price that each person would have to pay for the entire trip would be \$361.25.

1. Define a variable to represent the unknown.
 x represents number of people going on this trip.
2. Write an expression for the 3-day ticket rate per person plus the total rental cost.
 $299x + 871.50$
3. Write another expression for the total amount paid per person for the trip, assuming Marissa and Mykuria are able to get enough friends to come on the trip, so that the vacation rental house will be at maximum capacity.
 $361.25x$
4. If the rental house is at maximum capacity, the *two* expressions will be equivalent to each other. Set the *two* expressions equal to each other to create an equation, then solve your equation.
 $299x + 871.50 = 361.25x$
 $871.50 = 62.25x$
 $14 = x$
5. Interpret your solution in the context of the problem.
14 total people need to come on vacation in order for the price paid per person to be \$361.25.
6. Explain why the *two* expressions would NOT be equivalent if only 6 people attended the trip.
In the expression $361.25x$, the coefficient 361.25 represents the price each person would have to pay only if there is enough people to fill up the room. Since the room can fit over 6, the price per person will be greater than if the room were full.

Javier is in a bowling league. On Friday nights he bowls at Pinz Bowling Alley where it costs \$11.47 per game and another \$9.55 to rent bowling shoes. On Saturday nights he bowls at Vista Bowl where games cost \$12.50 each, shoe rentals cost \$3.95, and drinks cost \$1.48. This past Saturday at Vista Bowl, Javier bowled the same number of games he bowled the night before at Pinz. He also spent the same amount of money at Vista Bowl that he spent at Pinz, even though he also purchased one drink at Vista Bowl. How many games did Javier bowl at each bowling alley?

7. Define a variable to represent the unknown.

Answers may vary slightly.

x represents number of games Javier bowled

8. Write and solve an equation that represents this situation.

$$11.47x + 9.55 = 12.5x + 3.95 + 1.48$$

$$1.03x = 4.12$$

$$x = 4$$

9. Interpret your solution.

Javier bowled 4 games at each bowling alley.

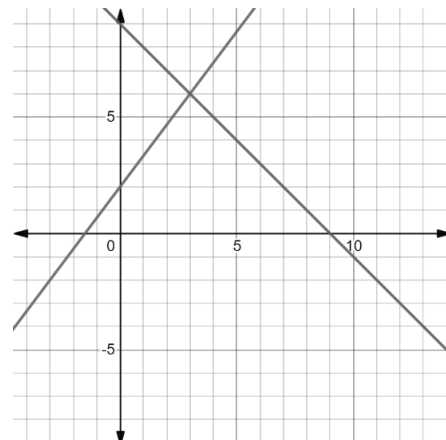
10. Why is 6.5 not a viable solution for this context?

Games bowled needs to be an integer since he can't pay for 0.5 games.

Algebra 1
Unit 5
Lesson 6 Homework

Name Answer Key
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The graph of the system of equations $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$ is shown.



1. To determine the solution to a system of equations, identify the

- ☐ slope.
- ☐ y-intercept.
- ☒ point of intersection.

2. Determine the solution to the system of equations from the graph.

$x = 3, y = 6$
 $(3, 6)$

Verify algebraically that that your solution satisfies both equations in the system.

3. $y = \frac{4}{3}x + 2$

$(6) = \frac{4}{3}(3) + 2$
 $6 = 4 + 2$
 $6 = 6 \checkmark$

4. $y = -x + 9$

$(6) = -(3) + 9$
 $6 = -3 + 9$
 $6 = 6 \checkmark$

5. Choose all of the systems of equations that are equivalent to the original system of equations, $\begin{cases} y = \frac{4}{3}x + 2 \\ y = -x + 9 \end{cases}$. Explain how you know the systems are equivalent.

$\begin{cases} -4x + 3y = 6 \\ x + y = 9 \end{cases}$	$\begin{cases} y = \frac{4}{3}x + 2 \\ 4x + 4y = 36 \end{cases}$	$\begin{cases} -\frac{4}{3}x + y = 2 \\ -x - y = -9 \end{cases}$	$\begin{cases} 8x - 6y = -2 \\ y = -x + 9 \end{cases}$
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After rearranging the equations, the slopes and y-intercepts match.

6. Verify your choices in question 5 algebraically.

$-4x + 3y = 6$
 $3y = 4x + 6$
 $y = \frac{4}{3}x + 2 \checkmark$

$x + y = 9$
 $y = -x + 9 \checkmark$

$4x + 4y = 36$
 $4y = -4x + 36$
 $y = -x + 9 \checkmark$

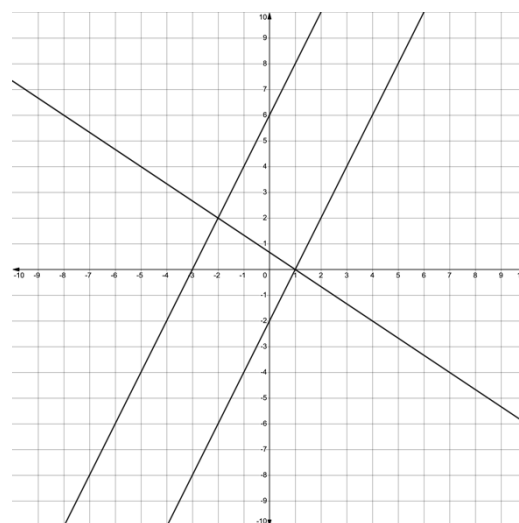
$-\frac{4}{3}x + y = 2$
 $y = \frac{4}{3}x + 2 \checkmark$

$-x - y = -9$
 $-y = x - 9$
 $y = -x + 9 \checkmark$

The graphs of the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ and $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ are shown.

7. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$ is $(\underline{1}, \underline{0})$. Verify the solution algebraically.

$$\begin{array}{rcl} 2x + 3y & = & 2 \\ -2x + y & = & -2 \\ \hline 4y & = & 0 \\ y & = & 0 \end{array} \quad \begin{array}{rcl} -2x + 0 & = & -2 \\ -2x & = & -2 \\ x & = & 1 \end{array}$$



8. The solution to the system $\begin{cases} 2x + 3y = 2 \\ -2x + y = 6 \end{cases}$ is $(\underline{-2}, \underline{2})$. Verify the solution algebraically.

$$\begin{array}{rcl} 2x + 3y & = & 2 \\ -2x + y & = & 6 \\ \hline 4y & = & 8 \\ y & = & 2 \end{array} \quad \begin{array}{rcl} 2x + 3(2) & = & 2 \\ 2x + 6 & = & 2 \\ 2x & = & -4 \\ x & = & -2 \end{array}$$

9. Fill in the blanks with words from the word bank to complete the sentence.

The two systems of equations are <u>not equivalent</u> because the two systems result in <u>different</u> solution(s).	Word Bank: ☐ equivalent ☐ equal ☒ not equivalent ☐ not equal ☐ the same ☒ different
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10. Replace one equation in the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = 5 \end{cases}$ so that it is equivalent, but not equal to the system of equations $\begin{cases} 2x + 3y = 2 \\ -2x + y = -2 \end{cases}$.

Answers may vary.

$$y = 2x - 2 \quad \begin{cases} 2x + 3y = 2 \\ y = 2x - 2 \end{cases}$$

Algebra 1

Unit 5

Lesson 7 Homework

Name Answer Key

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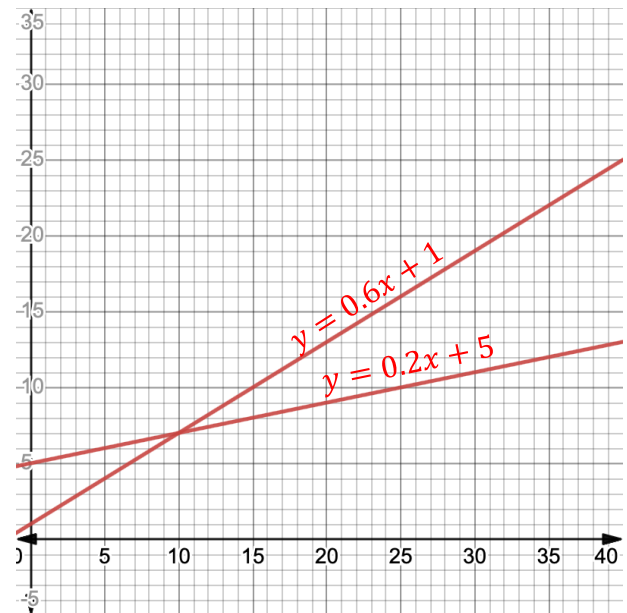
Period _____

Use the system of equations $\begin{cases} y = 0.2x + 5 \\ y = 0.6x + 1 \end{cases}$ to complete the following.

- Graph the system of equations on the coordinate plane.

- Complete the statement.

The point (10, 7) is the solution to the system.



Jake has a total of 12 quarters and dimes worth \$1.95. The system of equations $\begin{cases} d = 12 - q \\ d = 19.5 - 2.5q \end{cases}$ can be used to represent the amount of quarters, q , and the amount of dimes, d .

The graph of the system is shown.

- Label the axes.
- What are any possible constraints on the x -values?

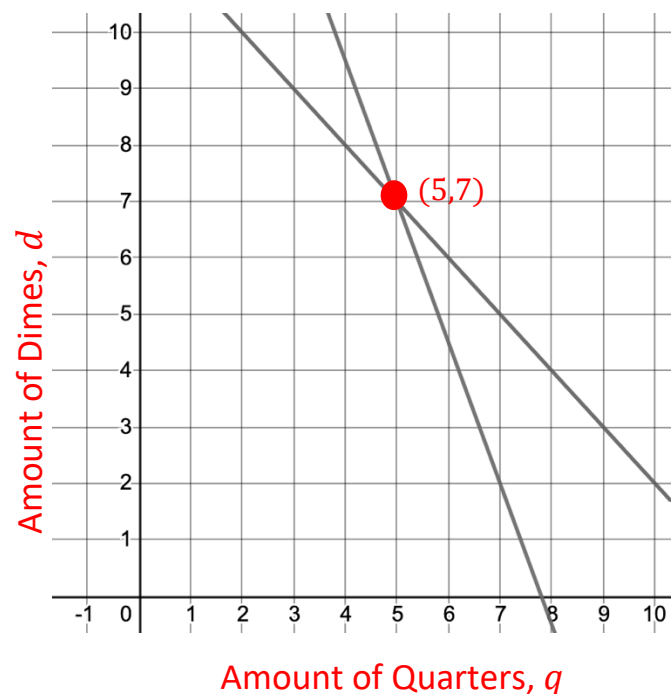
$$x \geq 0$$

- What are any possible constraints on the y -values?

$$y \geq 0$$

- What is the solution to this system of equations?

$$(5, 7)$$

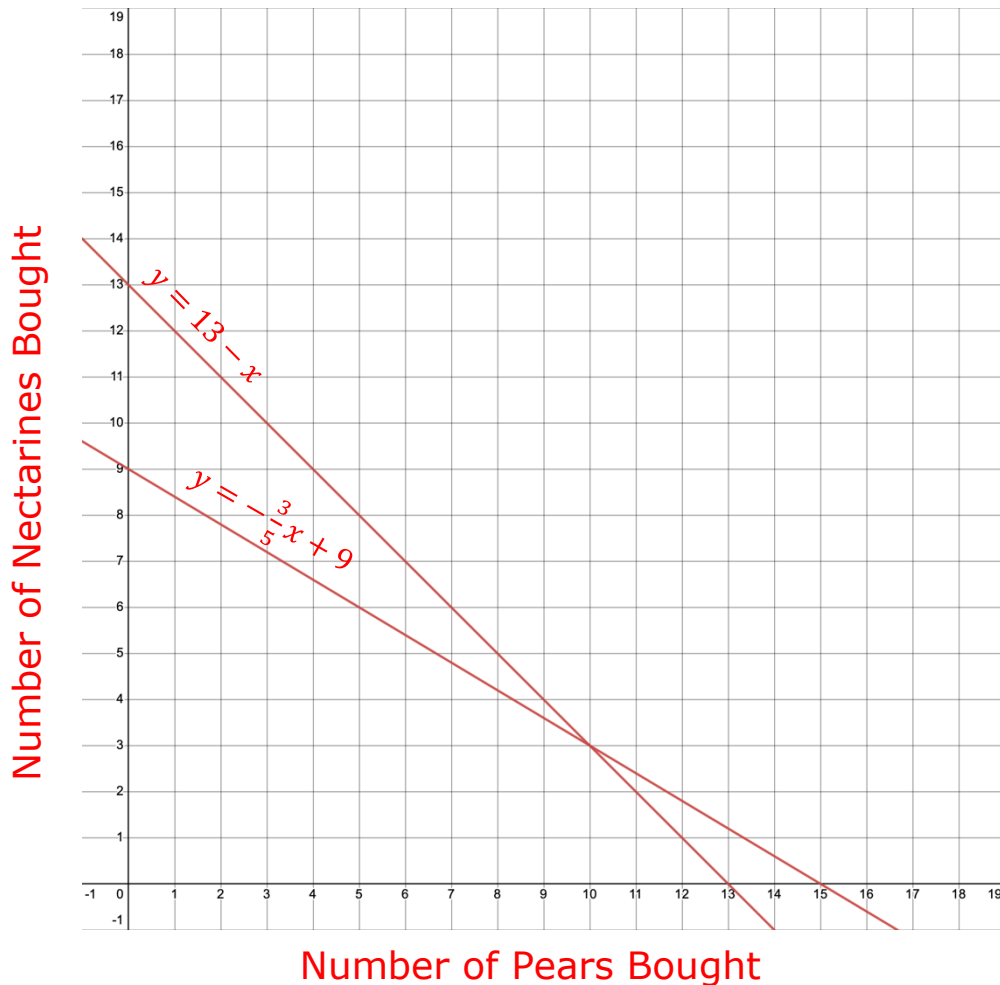


- Interpret the meaning of the solution in the context.

Jake has 5 quarters and 7 dimes.

Xander went to the grocery store and bought pears for \$0.75 each and nectarines for \$1.25 each. He bought a total of 13 pieces of fruit and spent \$11.25. The system of equations $\begin{cases} y = 13 - x \\ y = -\frac{3}{5}x + 9 \end{cases}$ can be used represents the situation where x is the number of pears and y is the number of nectarines bought.

8. Graph the system of equations. Label each axis with the fruit it represents.



9. What is the solution to the system?
(10,3)
10. Interpret the meaning of the solution in the context.
Xander bought 10 pears and 3 nectarines.

Algebra 1
Unit 5
Lesson 8 Homework

Name Answer Key
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Solve each system of equations using substitution for questions 1-4.

$$1. \begin{cases} 3x + y = -9 \\ x = 6y \end{cases} \quad \begin{aligned} 3(6y) + y &= -9 \\ 18y + y &= -9 \\ 19y &= -9 \\ y &= -\frac{9}{19} \end{aligned} \quad \begin{aligned} x &= 6\left(-\frac{9}{19}\right) \\ x &= -\frac{54}{19} \end{aligned}$$

$\left(-\frac{54}{19}, -\frac{9}{19}\right)$

$$2. \begin{cases} y = 2x \\ y = -2x + 12 \end{cases} \quad \begin{aligned} 2x &= -2x + 12 \\ 4x &= 12 \\ x &= 3 \end{aligned} \quad \begin{aligned} y &= 2(3) \\ y &= 6 \end{aligned}$$

$(3, 6)$

$$3. \begin{cases} 2x + 3y = 5 \\ x = -2y \end{cases} \quad \begin{aligned} 2(-2y) + 3y &= 5 \\ -4y + 3y &= 5 \\ -y &= 5 \\ y &= -5 \end{aligned} \quad \begin{aligned} x &= -2(-5) \\ x &= 10 \end{aligned}$$

$(10, -5)$

$$4. \begin{cases} 4y - 1 = x \\ 8x + 7y = -47 \end{cases} \quad \begin{aligned} 8(4y - 1) + 7y &= -47 \\ 32y - 8 + 7y &= -47 \\ 39y &= -39 \\ y &= -1 \end{aligned} \quad \begin{aligned} 4(-1) - 1 &= x \\ -5 &= x \end{aligned}$$

$(-5, -1)$

Use the system $\begin{cases} 3x - 2y = 11 \\ y = \frac{1}{2}x - 1 \end{cases}$ for questions 5 - 6.

5. Describe why using substitution is an efficient method to solving the system.
Substitution is an efficient method to solve a system of equations when one variable is already isolated or solved for in terms of the other variable.

6. Solve the system.

$$\begin{aligned} 3x - 2\left(\frac{1}{2}x - 1\right) &= 11 & y &= \frac{1}{2}\left(\frac{9}{2}\right) - 1 \\ 3x - x + 2 &= 11 & y &= \frac{9}{4} - \frac{4}{4} \\ 2x &= 9 & y &= \frac{5}{4} \\ x &= \frac{9}{2} \\ \left(\frac{9}{2}, \frac{5}{4}\right) &\text{ or } (4.5, 1.25) \end{aligned}$$

John and Max work at a sandwich shop. John can make 15 sandwiches per hour, and Max can make 10 sandwiches per hour. Max worked 5 more hours than John and they made a total of 150 sandwiches that day. Determine the number of hours Max worked and the number of hours John worked.

7. Name the *two* variables that are being related in this situation.

Let m represent the number of hours Max works.

Let j represent the number of hours John works.

8. Write a system of equations to represent the number of hours Max and John worked.

$$\begin{cases} 15j + 10m = 150 \\ m = j + 5 \end{cases}$$

9. Solve the system of equations using substitution.

$$15j + 10(j + 5) = 150$$

$$m = 4 + 5$$

$$15j + 10j + 50 = 150$$

$$m = 9$$

$$25j = 100$$

$$j = 4$$

10. Interpret the solution of the solution.

John worked for 4 hours, and Max worked for 9 hours to make a total of 150 sandwiches.

Algebra 1
Unit 5
Lesson 9 Homework

Name Answer Key
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1. Solve the system of equations using elimination.

$$\begin{array}{rcl}
 + \begin{cases} 3x + 6y = 18 \\ -3x + 4y = 2 \end{cases} & & \begin{array}{l} 3x + 6(2) = 18 \\ 3x + 12 = 18 \\ 3x = 6 \\ x = 2 \end{array} \\
 \hline & & (2,2) \\
 10y = 20 & & \\
 y = 2 & &
 \end{array}$$

2. Verify the solution from question 1 algebraically using substitution.

$$\begin{array}{ll}
 3(2) + 6(2) = 18 & -3(2) + 4(2) = 2 \\
 6 + 12 = 18 & -6 + 8 = 2 \\
 18 = 18 \checkmark & 2 = 2 \checkmark
 \end{array}$$

3. Solve the system of equations using elimination.

$$\begin{array}{rcl}
 + \begin{cases} 2x - 7y = 10 \\ 4x + 7y = 20 \end{cases} & & \begin{array}{l} 2(5) - 7y = 10 \\ 10 - 7y = 10 \\ -7y = 0 \\ y = 0 \end{array} \\
 \hline & & (5,0) \\
 6x = 30 & & \\
 x = 5 & &
 \end{array}$$

4. Verify the solution from question 3 algebraically using substitution.

$$\begin{array}{ll}
 2(5) - 7(0) = 10 & 4(5) + 7(0) = 20 \\
 10 = 10 \checkmark & 20 = 20 \checkmark
 \end{array}$$

5. Solve the system of equations using elimination.

$$\begin{array}{rcl}
 \begin{cases} 2x - 4y = -12 \\ x + 3y = 9 \end{cases} & \begin{array}{l} 2x - 4y = -12 \\ -2x - 6y = -18 \\ \hline -10y = -30 \\ y = 3 \end{array} & \begin{array}{l} 2x - 4(3) = -12 \\ 2x - 12 = -12 \\ 2x = 0 \\ x = 0 \end{array} \\
 & & (0,3)
 \end{array}$$

6. Verify the solution from question 5 algebraically using substitution.

$$\begin{array}{ll}
 2(0) - 4(3) = -12 & (0) + 3(3) = 9 \\
 -12 = -12 \checkmark & 9 = 9 \checkmark
 \end{array}$$

7. Solve the system of equations using elimination.

$$\begin{array}{rcl}
 \left(\begin{array}{l} 3x - 4y = -14 \\ 2x - 6y = 8 \end{array} \right) \begin{array}{l} \times 2 \\ -3 \end{array} & \begin{array}{l} 6x - 8y = -28 \\ -6x + 18y = -24 \end{array} & \begin{array}{l} 2x - 6(-5.2) = 8 \\ 2x + 31.2 = 8 \end{array} \\
 & \hline & \begin{array}{l} 10y = -52 \\ y = -5.2 \end{array} & \begin{array}{l} 2x = -23.2 \\ x = -11.6 \end{array} \quad (-11.6, -5.2)
 \end{array}$$

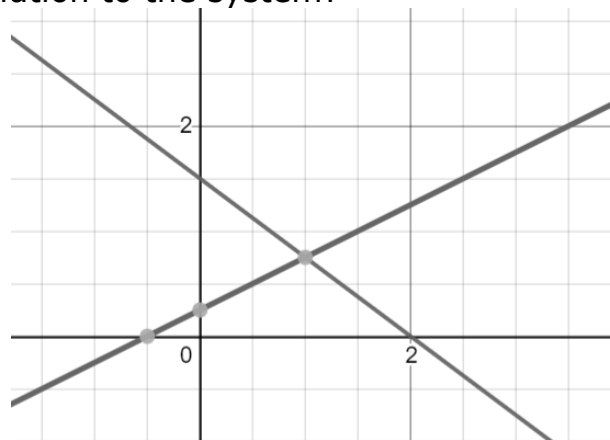
8. Verify the solution from question 7 algebraically using substitution.

$$\begin{array}{ll}
 3(-11.6) - 4(-5.2) = -14 & 2(-11.6) - 6(-5.2) = 8 \\
 -34.8 + 20.8 = -14 & -23.2 + 31.2 = 8 \\
 -14 = -14 \checkmark & 8 = 8 \checkmark
 \end{array}$$

9. Solve the system of equations using elimination.

$$\begin{array}{rcl}
 - \left\{ \begin{array}{l} 3x + 4y = 6 \\ -2x + 4y = 1 \end{array} \right. & \begin{array}{l} -2(1) + 4y = 1 \\ -2 + 4y = 1 \end{array} & \\
 \hline & \begin{array}{l} 4y = 3 \\ y = \frac{3}{4} \end{array} & \left(1, \frac{3}{4} \right) \\
 \begin{array}{l} 5x = 5 \\ x = 1 \end{array} & &
 \end{array}$$

10. The graph of the system from question 9 is shown below. How can the graph be used to verify the solution to the system?



Look at the point of intersection. Here, $\left(1, \frac{3}{4} \right)$ is the point of intersection, verifying the solution is correct.

Algebra 1
Unit 5
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

1. Solve the system of equations using elimination.

$$\begin{array}{rcl} \left(\begin{array}{l} 6y + 3x = -6 \\ 10x - 4y = 28 \end{array} \right) \begin{array}{l} 2 \\ 3 \end{array} & \begin{array}{l} 6x + 12y = -12 \\ +30x - 12y = 84 \\ \hline 36x = 72 \\ x = 2 \end{array} & \begin{array}{l} 6y + 3(2) = -6 \\ 6y + 6 = -6 \\ 6y = -12 \\ y = -2 \end{array} \end{array} \quad (2, -2)$$

2. Verify the solution to question 1 algebraically using substitution.

$$\begin{array}{ll} 6(-2) + 3(2) = -6 & 10(2) - 4(-2) = 28 \\ -12 + 6 = -6 & 20 + 8 = 28 \\ -6 = -6 \checkmark & 28 = 28 \checkmark \end{array}$$

Jasper is working two different jobs over the summer. He babysits for \$9 an hour and \$15 per hour mowing lawns. Last week Jasper made a total of \$144 and worked a combined total of 14 hours. This situation can be represented by the system of equations shown where b represents the hours he babysits and l represents the amount of hours he mowed lawns.

$$\begin{cases} 9b + 15l = 144 \\ b + l = 14 \end{cases}$$

3. The system could be solved using substitution or elimination. Which method do you prefer to use?

Answers may vary.
Elimination

4. Solve the system of equations using your preferred method.

$$\begin{array}{ll} 9b + 15l = 144 & b + l = 14 \\ -15b - 15l = -210 & 11 + l = 14 \\ \hline -6b = -66 & l = 3 \\ b = 11 & \end{array}$$

5. Interpret your solution.

Last week, Jasper babysat for 11 hours and mowed lawns for 3 hours.

6. Describe the constraints on b and l in this context.

b and l must be 0 or greater.
 $b \geq 0, l \geq 0$

Jackson and Miriam went to a restaurant and ordered one plain pizza and two sodas. Their bill totaled \$15.95. Later that day, *five* friends went to the same restaurant. They ordered *three* plain pizzas and each person had one soda. Their bill totaled \$45.90. This situation can be represented by the system of equations shown where p represents the cost of one plain pizza and s represents the cost of a soda.

$$\begin{cases} p + 2s = 15.95 \\ 3p + 5s = 45.90 \end{cases}$$

7. The system could be solved using both substitution or elimination. Which method do you prefer to use?

Answers may vary.

Elimination

8. Solve the system of equations.

$$\begin{array}{r} 3p + 6s = 47.85 \\ -3p - 5s = -45.90 \\ \hline s = 1.95 \end{array}$$

$$\begin{array}{r} p + 2(1.95) = 15.95 \\ p + 3.90 = 15.95 \\ p = 12.05 \end{array}$$

9. Interpret the solution.

A plain pizza cost \$12.05, and a soda costs \$1.95.

10. Determine the constraints on p and s in this context.

$$p > 0, s > 0$$

This is because the cost of a pizza and the cost of a soda must be more than \$0.00 each.

Rebecca has \$1.90 worth of nickels and dimes. She has 2 more nickels than dimes.

1. Define the variables to write the system of equations.

Let x = number of nickels

Let y = number of dimes

2. Write a system of equations to determine the number of nickels and dimes Rebecca has.

$$\begin{cases} 0.05x + 0.10y = 1.90 \\ x - 2 = y \end{cases}$$

3. Solve the system using the method of your choice.

$$\begin{aligned} 0.05x + 0.10(x - 2) &= 1.90 & y &= (14) - 2 \\ 0.05x + 0.1x - 0.2 &= 1.90 & y &= 12 \\ 0.15x &= 2.10 \\ x &= 14 \end{aligned}$$

4. Rebecca has 14 nickels and 12 dimes.

Sampson and Riki went to the movies to watch 'The Batman' that was just released. Sampson bought 2 bags of popcorn and 5 pretzels, costing a total of \$37.25. Riki bought 8 bags of popcorn and 9 pretzels, costing a total of \$102.25. Determine the price of each bag of popcorn and the price of each pretzel.

5. Sampson created the equation $2b + 5p = 37.25$. Create a second equation that could be used to find the price of each bag of popcorn and the price of each pretzel.

$$8b + 9p = 102.25$$

6. Solve the system using the method of your choice.

$$(2b + 5p = 37.25) \cdot 4 \rightarrow 8b + 20p = 149.00$$

$$\begin{aligned} 8b + 20p &= 149.00 \\ -8b - 9p &= -102.25 \\ \hline 11p &= 46.75 \\ p &= 4.25 \end{aligned}$$

$$\begin{aligned} 2b + 5(4.25) &= 37.25 \\ 2b + 21.25 &= 37.25 \\ 2b &= 16 \\ b &= 8 \end{aligned}$$

7. The price of each bag of popcorn is \$8. The price of each pretzel is \$4.25.

A new Marvel movie was released at the Imax theater. The Friday 8 P.M. showing sold a total of 150 tickets. A child ticket costs \$7.75 and an adult ticket costs \$10.25. The show generated a total of \$1,470.

8. Write a system of equations that could be used to find the number of child tickets that were sold and the number of adult tickets that were sold.

$$\begin{cases} x + y = 150 \\ 7.75x + 10.25y = 1,470 \end{cases}$$

9. Solve the system of equations you wrote.

$$x + y = 150 \rightarrow y = 150 - x$$

$$7.75x + 10.25(150 - x) = 1,470$$

$$7.75x + 1,537.5 - 10.25x = 1,470$$

$$-2.5x = -67.5$$

$$x = 27$$

$$y = 150 - (27)$$

$$y = 123$$

27 children tickets and 123 adult tickets were sold.

Dale wants to build a new deck for his house. The nails come in large boxes and small boxes. The large boxes each have 25 nails, and the small boxes hold 20 nails. Dale bought a total of 295 nails and a total of 13 boxes.

10. Write a system of equations to find how many boxes of small and large boxes of nails Dale bought.

$$\begin{cases} 25l + 20s = 295 \\ l + s = 13 \end{cases}$$

11. Solve the system of equations you wrote.

$$l + s = 13 \rightarrow l = 13 - s$$

$$25(13 - s) + 20s = 295$$

$$325 - 25s + 20s = 295$$

$$-5s = -30$$

$$s = 6$$

$$l = 13 - (6)$$

$$l = 7$$

Dale bought 7 large boxes and 6 small boxes of nails.